Project Development Phase
Model Performance Test

Date	10 February 2025
Team ID	LTVIP2026TMIDS80982
Project Name	Prosperity Prognosticator: Machine Learning for Startup Success Prediction
Maximum Marks	

Model Performance Testing:

(Project: Startup Success Prediction using Machine Learning)
Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1	Model Summary	Algorithm Used: Random Forest Classifier Number of Features: 10 (Funding Amount, Funding Rounds, Milestones, Relationships, etc.) Train-Test Split: 80% Training – 20% Testing Target Variable: Startup Outcome (Acquired / Closed)	Attach Model Training Output Screenshot
2	Accuracy	Training Accuracy – 92% Validation Accuracy – 87%	Attach Accuracy Output Screenshot
3	Fine Tuning Result (if Done)	Validation Accuracy after Hyperparameter Tuning – 90% (Using GridSearchCV for optimizing n_estimators, max_depth, etc.)	Attach Fine-Tuning Result Screenshot

Explanation:

- Training Accuracy represents how well the model performs on training data.
- Validation Accuracy represents how well the model generalizes on unseen data.
- Fine-tuning improves model performance by adjusting hyperparameters of the Random Forest model.